



FACULTY OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

CSM3601-Problem Solving Using Artificial Intelligence

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PROJECT PROPOSAL

STOCK PRICE PREDICTION

PREPARED FOR:

Assoc. Prof. Dr Azreen Azman

PREPARED BY: GROUP 13

NO.	NAME	MATRIC NUMBER
1.	BAIYIXIONG	226593
2.	ZHUMAILUN	227233
3.	DAIMINGYI	226662
4.	WUHONGYU	227078

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1.Project title

Stock Price Prediction Using Machine Learning

2.Problem statement

Stock price forecasting is a typical real-world problem that matches the supervised learning (regression) requirements of CSM3601: Problem Solving Using Artificial Intelligence. For AAPL (Apple Inc.) stock, the core problems to be solved are as follows:

1. AAPL stock price presents strong nonlinearity and time-series volatility, which cannot be effectively fitted by traditional statistical methods. It is necessary to use the machine learning regression algorithms learned in class to mine hidden patterns.
2. Real-world stock data contain massive historical information, but it lacks systematic processing via the standard ML workflow taught in class (including data cleaning, EDA, feature engineering) to extract effective features that affect closing prices.
3. Single machine learning models often have limited prediction performance.
According to course requirements, it is necessary to compare multiple algorithms and optimize hyperparameters, but there is no complete comparison scheme for AAPL stock price prediction.
4. Qualitative trend judgment is not intuitive enough. As required by the course, we need to use regression evaluation metrics (MAE, MSE, RMSE, R^2) to quantitatively predict specific closing prices and build a complete, interpretable machine learning pipeline.

3.Objectives

1. To apply the ****complete supervised machine learning (regression) workflow**** taught in CSM3601 to collect, clean, preprocess and conduct exploratory data analysis (EDA) on AAPL historical stock data.

2. To perform **feature engineering** as learned in class, construct effective trading features and extract key information that drives AAPL stock price changes.
3. To develop, train and compare **at least 3 machine learning regression models** (Linear Regression, Random Forest Regressor, XGBoost Regressor) and optimize hyperparameters to avoid overfitting.
4. To evaluate model performance using the **standard regression metrics** required by the course (MAE, MSE, RMSE, R^2) and determine the best-performing model.
5. To interpret prediction results, analyze feature importance and demonstrate critical thinking in model selection, so as to complete the whole process of machine learning solution design and result communication.

4. Proposed Dataset

For this project, we propose to use a historical stock price dataset to predict future stock prices. The dataset will be collected from public financial data sources such as Kaggle or Yahoo Finance. These platforms provide structured stock market data that can be used for machine learning analysis.

The dataset will focus on one selected company stock, such as Apple, Tesla, Microsoft, or another actively traded company. Focusing on one company makes the prediction task more specific and manageable for this project.

The proposed dataset will include important stock market attributes such as Date, Open Price, High Price, Low Price, Close Price, Adjusted Close Price, and Trading Volume. The Close Price will be used as the main target variable because it represents the final trading price of the stock at the end of each day. Other attributes such as Open, High, Low, Adjusted Close, and Volume will be used as input features to help the model learn historical price patterns.

Before developing the machine learning models, the dataset will be cleaned and preprocessed. Missing values will be checked and handled, unnecessary columns may be removed, and the data will be arranged based on date order. Feature engineering will also be applied by creating additional features such as daily price change, daily return, moving average, and volatility. These features can help improve the model's ability to understand stock market trends.

Overall, this dataset is suitable for the project because it is numerical, structured, publicly available, and directly related to the problem of stock price prediction. It also supports the use of regression-based machine learning models for predicting future closing prices.

5. Proposed algorithms

To accurately predict the closing price and short-term trend of Apple Inc. (AAPL) stock, this project will employ various time series forecasting and machine learning algorithms, selecting the optimal model through comparative experiments. The main algorithms include:

1. Long Short-Term Memory Network (LSTM)**

As a variant of Recurrent Neural Network (RNN), LSTM effectively captures long-term dependencies in stock prices, avoiding the vanishing gradient problem. We will construct a multi-layer LSTM model, taking historical prices, trading volumes, and other time-series features as input, and outputting the predicted price for the next trading day.

2. Gated Recurrent Unit (GRU)**

GRU has a simpler structure and faster training speed than LSTM, performing comparably in certain sequence tasks. It will be used as one of the comparison models to verify its ability to fit AAPL price fluctuations.

3. Autoregressive Integral Moving Average (ARIMA)**

As a benchmark for traditional statistical models, ARIMA is used to capture linear trends and seasonality. The optimal parameters (p, d, q) are determined through the ACF/PACF plot, providing a comparison with deep learning models.

4. XGBoost (optional comparison) extracts features such as past 5-day returns, moving averages, and the Relative Strength Index (RSI), and uses gradient boosting trees for regression prediction to evaluate the performance of non-time-series models.

Model evaluation metrics: Mean Squared Error (MSE), Mean Absolute Error (MAE), Coefficient of Determination (R^2), and Directional Accuracy.

6. Expected Outcome

The expected outcome of this project is to successfully develop a machine learning-based stock price prediction system that can forecast the future closing price of AAPL stock with acceptable accuracy. By applying the complete supervised learning workflow, the project is expected to demonstrate how machine learning techniques can be used to solve real-world financial prediction problems.

First, the dataset is expected to be successfully cleaned, processed, and transformed into a suitable format for machine learning analysis. Through exploratory data analysis (EDA), important patterns and trends such as stock price fluctuations, trading volume changes, and moving average behaviors are expected to be identified.

Second, feature engineering is expected to improve the prediction capability of the models. Features such as daily return, moving averages, volatility, and lagged prices are expected to provide useful information that helps the algorithms better understand historical stock price movements.

Third, multiple regression and time-series prediction models, including Linear Regression, Random Forest, XGBoost, LSTM, GRU, and ARIMA, are expected to produce different levels of prediction accuracy. Deep learning models such as LSTM and GRU are expected to perform better in capturing nonlinear and sequential patterns in stock prices compared to traditional regression models.

Furthermore, the project is expected to identify the best-performing model based on evaluation metrics such as MAE, MSE, RMSE, R^2 , and Directional Accuracy. The selected model is expected to achieve lower prediction errors and provide more reliable future stock price forecasts.

Finally, the project is expected to produce visualizations and analytical results such as predicted vs actual stock price graphs, trend analysis, and feature importance analysis. These results will help users better understand stock market behavior and demonstrate the practical application of artificial intelligence and machine learning techniques in financial forecasting.